

KG+LLM Task 2: Hybrid GraphRAG for Public Administration

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Abstract

Public Administration (PA) regulations are characterized by high complexity, strict terminological precision, and interconnected legal frameworks. Standard Retrieval-Augmented Generation (RAG) systems, relying solely on vector similarity, often fail to capture the structural and hierarchical logic inherent in legal texts (e.g., roles, duties, and mandates). This project presents a Hybrid GraphRAG system applied to the *Empulia* regulations (Regione Puglia). By integrating a Knowledge Graph (KG) constructed via the **Qwen3-14B** Large Language Model with vector-based retrieval, the system achieves superior factual accuracy and reasoning capabilities. We introduce a novel pipeline featuring semantic entity resolution to handle graph sparsity and demonstrate, through a real-world case study on procurement laws, that the hybrid approach successfully bridges the gap between specific regional entities and national regulatory frameworks.

Keywords

Knowledge Graphs, Large Language Models, Retrieval-Augmented Generation, Legal NLP, Public Administration, Qwen3

1. Introduction and Motivations

The integration of Large Language Models (LLMs) into the Public Administration (PA) domain offers significant potential for simplifying access to complex regulatory frameworks. However, LLMs suffer from hallucinations and a lack of up-to-date knowledge [1]. While Retrieval-Augmented Generation (RAG) addresses this by retrieving relevant text chunks, standard vector-based RAG struggles with "global reasoning" tasks that require connecting disparate pieces of information across different documents [2].

In the context of the *Empulia* platform (the central purchasing body for Regione Puglia), a user might ask: "What are the duties of *InnovaPuglia*?". The answer lies in a logical chain: *InnovaPuglia* is designated as a *Soggetto Aggregatore* (Regional Law), and the duties of a *Soggetto Aggregatore* are defined in a separate National Decree. A vector-only system might retrieve the definition of the role but fail to explicitly link it to the specific entity.

To address this, we propose a **Hybrid GraphRAG** architecture. By constructing a structured Knowledge Graph (KG) that maps entities (e.g., Ente, Ruolo) and their relationships (e.g., RIVESTE_RUOLO), we enable the LLM to perform multi-hop reasoning. This project aims to demonstrate that combining structured knowledge (KG) with unstructured text (Vector Search) significantly improves factual accuracy and explainability in the legal domain.

2. Related Work

Recent literature highlights the limitations of purely vector-based retrieval in complex domains. Edge et al. [2] introduced *GraphRAG*, demonstrating that graph-based indices outperform baseline RAG in sense-making tasks over large corpora, particularly when the answer requires synthesizing information from multiple sources. Our work adopts this paradigm but tailors it to the legal domain by enforcing a strict ontology, addressing the "hallucination" risks identified by Agrawal et al. [1].

Pan et al. [3] argue for the unification of LLMs and KGs, where KGs provide the structural grounding that LLMs lack. We implement this unification by using the KG not just for retrieval, but for "reasoning

support"—the LLM uses the graph to infer relationships (e.g., if A is B, and B does C, then A does C).

Furthermore, benchmarking studies by Ahmad et al. [4] indicate that hybrid approaches (Graph + Vector) achieve higher context relevance compared to traditional methods. However, most existing approaches do not address the specific challenge of "Graph Sparsity" in automated extraction. Our work contributes to this area by introducing a Semantic Entity Resolution pipeline that uses vector embeddings to merge disconnected nodes (e.g., merging "Consip" and "Consip S.p.A."), a critical step often overlooked in standard implementations.

3. Proposed Approach

3.1. Dataset and Ontology

The dataset consists of 16 PDF documents representing the regulatory framework of *Empulia*, including Regional Laws (e.g., L.R. 37/2014) and National Decrees (e.g., D.P.R. 207/2010).

To ensure high-quality extraction and prevent LLM hallucinations, it was defined a strict Ontology (Figure 1). The schema defines specific Nodes and Relationships to capture the legal logic:

- **Nodes (Entities):**

- *Ente* (Institution, e.g., Regione Puglia)
- *Legge* (Law source)
- *Articolo* (Specific legal article)
- *Ruolo* (Role, e.g., Soggetto Aggregatore)
- *Dovere* (Duty/Task)
- *Procedura* (Administrative Procedure)
- *Documento* (Legal Document, e.g., Bando di Gara)

- **Relationships (Logic):**

- SANCISCE (*Legge* → *Articolo*): Maps the source law to its articles.
- RIVESTE_RUOLO (*Ente* → *Ruolo*): Critical for reasoning. Captures designations (e.g., "InnovaPuglia is the Subject Aggregator").
- HA_COMPITO (*Ruolo* → *Dovere*): Links a role to its specific duties.
- REGOLA (*Articolo* → All): The central connector indicating that a specific legal article regulates a Role, Duty, Procedure, or Document.
- DIPENDE_DA (*Ente* → *Ente*): Captures hierarchical or functional dependencies.



Figure 1: The Ontology Schema generated by Neo4j. It visualizes the strict structure imposed on the LLM, where *Articolo* acts as the central normative hub connecting *Ruoli*, *Doveri*, *Procedure*, and *Documenti*.

3.2. Architecture and Implementation

The system was developed using a hybrid environment to optimize computational resources, decoupling the computationally expensive graph construction from the real-time inference.

3.2.1. Phase 1: KG Construction (Kaggle GPU)

We utilized a Kaggle notebook equipped with T4 GPUs to perform the extraction. The total execution time for the Knowledge Graph creation was 1 hour, 45 minutes, and 35 seconds.

- **Environment & Tools:** We used llama-cpp-python compiled with CUDA support for GPU acceleration. The database connection was managed via neo4j driver connecting to a Neo4j AuraDB cloud instance.
- **Data Ingestion Strategy:** To ensure fairness and deployability, we adopted a "Full Ingestion" strategy rather than filtering by keywords. We used RecursiveCharacterTextSplitter with a chunk size of 2000 characters to respect the logical boundaries of legal articles. Each chunk was tagged with metadata containing the source law filename to enable precise citations during inference.
- **Model Configuration:** We employed Qwen3-14B (Quantized Q4_K_M). This model was selected for its state-of-the-art instruction-following capabilities. We configured the model with a context window of 32,768 tokens and a temperature of 0.0 to ensure deterministic and factual extraction.
- **Strict Extraction Pipeline:** To mitigate hallucinations, we implemented a robust extraction loop:

1. **System Prompt:** We utilized the `/no_think` mode to suppress internal monologue and forced the model to output triples in a strict pipe-separated format (SUBJECT | TYPE | RELATION | OBJECT | TYPE).
2. **Python-side Validation:** A sanitation layer filtered the model's output. We rejected nodes that did not match the allowed Ontology types (e.g., if the model invented a type "Sentence") or if the node ID was longer than 100 characters (indicating a hallucinated sentence rather than an entity).
- **Semantic Entity Resolution:** A major challenge in KGs is sparsity (e.g., "Consip" and "Consip S.p.A." treated as different nodes). We implemented a custom **Hybrid Resolution Module**:
 1. **Semantic Check:** We generated vector embeddings for all node names using sentence-transformers and calculated Cosine Similarity.
 2. **Lexical Check:** We applied FuzzyWuzzy to detect spelling variations.
 3. **Safety Valve:** We implemented a Regex-based safety check to prevent the merging of numbered laws (e.g., ensuring "Legge 10" and "Legge 11" remain distinct).

Nodes were merged only if they satisfied both high semantic similarity (> 0.85) and lexical overlap.

3.2.2. Phase 2: Hybrid Inference (Local Environment)

The inference engine runs locally, connecting to the pre-populated Neo4j AuraDB (Cloud).

- **Model:** For local inference, we utilized Qwen3-8B via LM Studio to ensure low-latency responses suitable for a real-time user interface.
- **Hybrid Retriever:** The system processes a user query in two parallel streams:
 1. **Vector Search:** Retrieves top- k text chunks using Cosine Similarity (captures context and definitions).
 2. **Graph Search:** Extracts entities from the query (e.g., "InnovaPuglia") and traverses the graph to find connected Roles and Duties via the RIVESTE_RUOLO and HA_COMPITO relationships.
- **User Interface:** A Streamlit app visualizes the answer, the reasoning process (Chain of Thought), and the retrieved graph triples for explainability.

4. Evaluation and Results

We evaluated the system using a "Batch Testing Suite" on domain-specific questions. We analyzed the contribution of the Graph versus the Vector store.

4.1. Vector vs. Graph Performance Analysis

To demonstrate the necessity of the hybrid approach, we analyzed the similarity scores (*check_scores.py*) for the query: *"Quali sono i compiti di InnovaPuglia come soggetto aggregatore?"*.

- **Result 1 (Score 0.84):** *Direttiva 2014/25/UE*. The vector model retrieved a European Directive about "innovazione", likely misled by the semantic similarity with the name "InnovaPuglia".
- **Result 2 (Score 0.83):** *L.R. 37/2014*. The correct regional law was ranked second.

This finding highlights a critical weakness of Vector Search: it can be "distracted" by semantic overlaps (Innovation vs. InnovaPuglia). The Knowledge Graph corrected this by providing the explicit, rigid link: *InnovaPuglia -[RIVESTE_RUOLO]-> Soggetto Aggregatore*, ensuring the LLM focused on the correct legal role rather than the generic concept of innovation.

4.2. Knowledge Graph Quality

The extraction process successfully mapped the complex relationships of the Empulia ecosystem. Figure 2 shows the ego-network of the *InnovaPuglia* node. The system correctly identified that InnovaPuglia:

- Holds the role of *Soggetto Aggregatore* (RIVESTE_RUOLO).
- Acts as a *Centrale di Committenza*.
- Is connected to the *Regione Puglia* via dependency relationships.



Figure 2: Graph visualization of the "InnovaPuglia" node. The system autonomously extracted relationships linking the specific regional entity to abstract legal roles like "Soggetto Aggregatore" and "Centrale di Committenza".

4.3. Inference

To evaluate the system's ability to integrate national and regional knowledge, we tested the query: "*Quali sono i compiti di Consip secondo la normativa?*".

This query is challenging because Consip is a national entity, but in the context of *Empulia*, it has specific relationships with regional bodies.

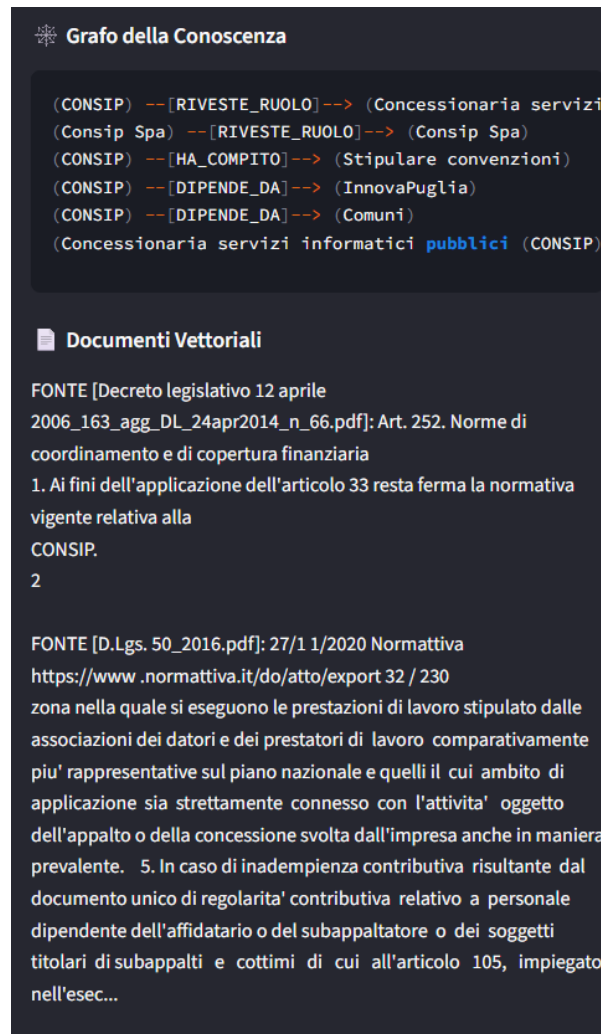


Figure 3: Retrieval Phase: The "Explainability Engine" (sidebar) shows the retrieved context. The Knowledge Graph (top) successfully identified the relationship (CONSIP) - [HA_COMPITO] -> (Stipulare convenzioni) and its dependency on *InnovaPuglia*. The Vector Search (bottom) retrieved the relevant Article 252 from D.Lgs. 163/2006.

As shown in Figure 3, the Hybrid Retriever successfully combined two distinct data sources:

- **Structured Logic (Graph):** It retrieved that Consip "Depends on InnovaPuglia" and has the task of "Stipulating conventions". This is a high-level architectural fact.
- **Unstructured Detail (Vector):** It retrieved the specific legal text of Art. 252, which defines the financial coverage for Consip's activities.

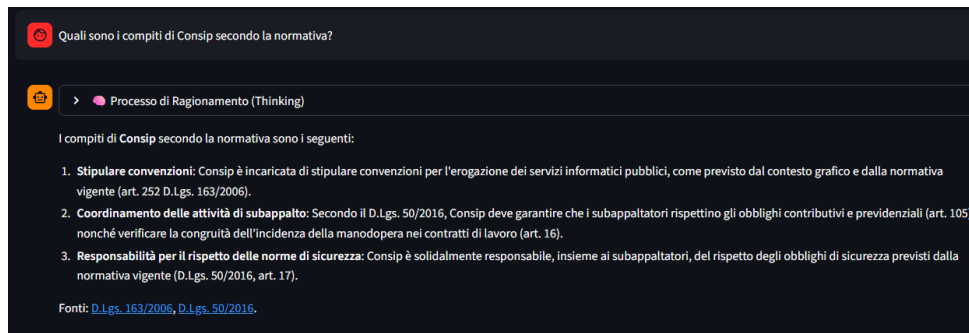


Figure 4: Reasoning Phase: The LLM processes the retrieved hybrid context. The "Thinking Process" (collapsed) shows the model synthesizing the graph's structural information with the vector's legal text to produce a final, cited answer.

Figure 4 demonstrates the final output. The model correctly synthesized the information, listing "Stipulare convenzioni" as a primary duty (derived from the Graph) and citing the specific Legislative Decree (derived from the Vector text), thus achieving the project's goal of factual accuracy and explainability.

5. Reproducibility

To ensure the reproducibility of our results, we detail the specific hyperparameters and technical configurations used in both the construction and inference phases.

Table 1
Technical Hyperparameters and Configuration

Parameter	Value
<i>Data Ingestion</i>	
Chunk Size	2000 characters
Chunk Overlap	200 characters
Embedding Model	sentence-transformers/all-MiniLM-L6-v2
<i>KG Construction (Kaggle)</i>	
Model	Qwen3-14B-Instruct (Quantized Q4_K_M)
Temperature	0.0 (Deterministic)
Context Window	32,768 tokens
Entity Resolution Threshold	Cosine Sim > 0.85 AND Fuzzy > 80
<i>Inference (Local)</i>	
Model	Qwen3-8B-Instruct (via LM Studio)
Temperature	0.6
Retrieval Strategy	Hybrid ($k = 2$ Vector + Graph Traversal)
Hardware	NVIDIA T4 (Kaggle) / Consumer CPU (Local)

The source code, including the extraction pipeline and the Streamlit interface, is available in the supplementary material.

6. Conclusions

We successfully developed a Hybrid GraphRAG system for Public Administration. The integration of Qwen3-14B for extraction and Neo4j for storage allowed us to bridge the gap between unstructured legal text and structured logic.

Key Findings:

1. Graph is essential for Logic: Vector search alone is susceptible to semantic drift (e.g., retrieving generic EU directives on innovation). The KG acts as a logical anchor, linking specific entities to their legal roles via `RIVESTE_RUOLO`.
2. Entity Resolution is critical: Without the semantic merging step, variations like "Consip Spa" and "CONSIP" would remain disconnected, breaking the reasoning chain.

7. Future Work and Limitations

While the proposed Hybrid GraphRAG system demonstrates significant qualitative improvements in reasoning and explainability for Public Administration tasks, several limitations remain that future research should address.

7.1. Quantitative Evaluation and Gold Standard

The primary limitation of this study is the absence of rigorous quantitative metrics such as Precision, Recall, F1 for the retrieval component. Currently, the evaluation relies on qualitative case studies and similarity scores. This is due to the lack of a pre-existing "Gold Standard" dataset (Ground Truth) for the specific domain of *Regione Puglia* regulations.

Future work will focus on the creation of an annotated evaluation dataset, developed in collaboration with legal experts, consisting of (Query, Context, Answer) triplets. This would allow for the application of automated evaluation frameworks such as **RAGAS** (Retrieval Augmented Generation Assessment) to mathematically measure:

- **Context Precision:** The proportion of relevant chunks retrieved from the Vector Store.
- **Graph Recall:** The ability of the Knowledge Graph to traverse and retrieve all necessary logical links (e.g., ensuring no intermediate role is missed in a chain of responsibility).
- **Faithfulness:** Measuring how accurately the generated answer adheres to the retrieved legal text without hallucination.

7.2. Scalability and Incremental Updates

The current KG construction pipeline, while accurate, is computationally intensive and sometimes the Nodes contains **typos** (approx. 1.75 hours for initial ingestion on T4 GPUs). In a real-world PA scenario, regulations are frequently updated. Future implementations should explore **Incremental Knowledge Graph Construction**, where only new or modified laws are processed and merged into the existing graph, rather than rebuilding the entire structure from scratch.

7.3. Automated Verification

To further mitigate the risk of subtle legal hallucinations, a "Critic-Verifier" architecture could be introduced. This would involve a secondary LLM agent tasked solely with verifying the triples extracted by the primary model against the source text before they are committed to the Neo4j database, ensuring a higher degree of reliability for critical administrative data.

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